

HDOC - Day 29

Name- Aaryan Pratap Singh

Sap Id- 590032842

Batch - 15

Algorithm

1. Start.
2. Display the menu:
 - Count even, odd, positive, negative, and zero elements.
 - Search for an element using linear search.
 - Merge two arrays.
 - Exit.
3. Read the user's choice.
4. Use `switch(choice)`:
 - **Choice 1:** Read the array size and elements. Traverse the array and count even, odd, positive, negative, and zero elements.
 - **Choice 2:** Read the array and the element to search. Traverse the array using linear search. Display its position if found; otherwise display "Element not found".
 - **Choice 3:** Read two arrays. Copy the elements of both arrays into a third array and display the merged array.
 - **Choice 4:** Exit.
 - Otherwise, display an invalid-choice message.
5. Repeat the menu until the user selects Exit.
6. Stop.

Test-Case Table

Test Case	Operation	Input	Expected Output
1	Count elements	5 → 2 -3 0 4 -6	Even = 4, Odd = 1, Positive = 2, Negative = 2, Zero = 1
2	Count elements	4 → 1 3 5 7	Even = 0, Odd = 4, Positive = 4, Negative = 0, Zero = 0
3	Linear search	5 → 10 20 30 40 50, search 30	Element found at position 3
4	Linear	4 → 5 10 15 20, search	Element not found
5	Merge	1 2 3 and 4 5	Merged array: 1 2 3 4 5
6	Invalid size	Size = 0	Ask for a positive size again
7	Invalid	Choice = 7	Invalid choice
8	Exit	Choice = 4	Program terminated

C Program

```

#include <stdio.h>

int main()
{
    int choice;

    do
    {
        printf("\n==== HDOC DAY-29 =====\n");
        printf("1. Count even, odd, positive, negative and
zero elements\n");
        printf("2. Search an element using linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
            {
                int n, i;
                int even = 0, odd = 0;
                int positive = 0, negative = 0, zero = 0;

                do
                {
                    printf("Enter the number of elements: ");
                    scanf("%d", &n);

                    if (n <= 0)
                        printf("Size must be positive. Enter
again.\n");

                } while (n <= 0);

                int a[n];

                printf("Enter %d elements:\n", n);

                for (i = 0; i < n; i++)
                {
                    scanf("%d", &a[i]);
                }
            }
        }
    }
}

```

```

for (i = 0; i < n; i++)
{
    /* Count even and odd */
    if (a[i] % 2 == 0)
        even++;
    else
        odd++;

    /* Count positive, negative and zero */
    if (a[i] > 0)
        positive++;
    else if (a[i] < 0)
        negative++;
    else
        zero++;
}

printf("\nEven elements      = %d\n", even);
printf("Odd elements        = %d\n", odd);
printf("Positive elements = %d\n", positive);
printf("Negative elements = %d\n", negative);
printf("Zero elements      = %d\n", zero);

break;
}

case 2:
{
    int n, i, key;
    int position = -1;

    do
    {
        printf("Enter the number of elements: ");
        scanf("%d", &n);

        if (n <= 0)
            printf("Size must be positive. Enter
again.\n");

    } while (n <= 0);

    int a[n];

```

```

printf("Enter %d elements:\n", n);

for (i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}

printf("Enter the element to search: ");
scanf("%d", &key);

/* Linear Search */
for (i = 0; i < n; i++)
{
    if (a[i] == key)
    {
        position = i;
        break;
    }
}

if (position != -1)
{
    printf("Element %d found at position %d.
\n",
        key, position + 1);
}
else
{
    printf("Element %d not found in the
array.\n", key);
}

break;
}

case 3:
{
    int n1, n2, i;

    do
    {
        printf("Enter size of first array: ");
        scanf("%d", &n1);

        if (n1 <= 0)

```

```

        printf("Size must be positive. Enter
again.\n");

    } while (n1 <= 0);

    do
    {
        printf("Enter size of second array: ");
        scanf("%d", &n2);

        if (n2 <= 0)
            printf("Size must be positive. Enter
again.\n");

    } while (n2 <= 0);

    int a[n1], b[n2], merged[n1 + n2];

    printf("Enter %d elements of first array:\n",
n1);

    for (i = 0; i < n1; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter %d elements of second array:
\n", n2);

    for (i = 0; i < n2; i++)
    {
        scanf("%d", &b[i]);
    }

    /* Copy first array */
    for (i = 0; i < n1; i++)
    {
        merged[i] = a[i];
    }

    /* Copy second array */
    for (i = 0; i < n2; i++)
    {
        merged[n1 + i] = b[i];
    }

```

```

        printf("Merged array:\n");

        for (i = 0; i < n1 + n2; i++)
        {
            printf("%d ", merged[i]);
        }

        printf("\n");

        break;
    }

    case 4:
        printf("Program terminated.\n");
        break;

    default:
        printf("Invalid choice. Please enter a choice
from 1 to 4.\n");
    }

    } while (choice != 4);

    return 0;
}

```

Execution 1

```

[aaryanrajput@aaryans-macbook ~ % nano day30.c
[aaryanrajput@aaryans-macbook ~ % clang day30.c -o day30.c
[aaryanrajput@aaryans-macbook ~ % ./day30.c

===== HD0C DAY-29 =====
1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 1
Enter the number of elements: 7
Enter 7 elements:
-1 -3 2 4 0 5 9

Even elements      = 3
Odd elements       = 4
Positive elements  = 4
Negative elements  = 2
Zero elements      = 1

```

Execution 2

```
===== HDOC DAY-29 =====
1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Enter the element to search: 30
Element 30 found at position 3.
```

Execution 3

```
===== HDOC DAY-29 =====
1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter size of first array: 3
Enter size of second array: 4
Enter 3 elements of first array:
10 20 30
Enter 4 elements of second array:
40 50 60 70
Merged array:
10 20 30 40 50 60 70
```

Execution 4 - Exit

```
===== HDOC DAY-29 =====
1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 4
Program terminated.
aaryanrajput@aaryans-macbook ~ %
```